

**PROGRAMME** : Diploma Programme in Information Technology/Computer Technology

**COURSE** : Client Side Scripting (CSS)

**COURSE CODE** : 232213

**LEARNING AND ASSESSMENT SCHEME:**

| Learning Scheme          |    |    |     |     |                | Credits | Assessment Scheme |       |       |       |                     |     |     |     |                        |     | Total Marks |
|--------------------------|----|----|-----|-----|----------------|---------|-------------------|-------|-------|-------|---------------------|-----|-----|-----|------------------------|-----|-------------|
| Actual Contact Hrs./Week |    |    | SLH | NLH | Paper Duration |         | Theory            |       |       |       | Based on LL and TSL |     |     |     | Based on Self Learning |     |             |
|                          |    |    |     |     |                |         |                   |       |       |       | Practical           |     |     |     | SLA                    |     |             |
| CL                       | TL | LL |     |     |                |         | FA-TH             | SA-TH | Total | FA-PR | SA-PR               |     | SLA |     |                        |     |             |
| 00                       | 00 | 04 | 00  | 04  | 02             | --      | Max               | Max   | Max   | Min   | Max                 | Min | Max | Min | Max                    | Min | 25          |
|                          |    |    |     |     |                |         | --                | --    | --    | --    | 25                  | 10  | --  | --  | --                     | --  |             |

IKS Content: Total Learning Hours for Term: 00 Hrs

# Indicates Online Examination, @ Internal assessment

**1.0 RATIONALE:**

Learning client-side scripting with JavaScript is crucial because it empowers developers to create dynamic, interactive web applications that enhance user experience. JavaScript enables real-time content updates, form validations, and user-driven interactions without needing constant server communication, leading to faster and more responsive interfaces. Mastering JavaScript also opens doors to various frameworks and libraries, such as React and Angular, which are essential for modern web development. Moreover, understanding client-side scripting helps in optimizing performance and improving usability, as it allows developers to address and debug client-specific issues effectively. This knowledge is fundamental for building contemporary, user-centric web applications.

**2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:**

The aim of the course is to help the student to attain the following industry /employer expected outcome: students will be well-prepared for careers in software development, with a strong foundation in building scalable, performant, and maintainable web applications.

**3.0 COURSE OUTCOMES:**

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Acquires the knowledge of basics in Javascript and create interactive web pages.
2. Implement Arrays, Functions and String manipulation in JavaScript.
3. Develop the dynamic generated web content by using different form elements and events.
4. Use JavaScript to work with cookies and perform different operations on browser window.
5. Create webpages in JavaScript to implement menus.

**4.0 COURSE DETAILS:**

| Unit                                                     | Major Learning Outcomes<br>(in cognitive domain)                                                                                                                                            | Topics and Sub-topics                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit-I<br/>Introduction to JavaScript programming</b> | 1a.State features of JavaScript.<br>1b.Develop JavaScript to implement conditional statement for a given problem.<br>1c.Develop JavaScript to implement loop statement for a given problem. | 1.1 Using Java Script in HTML pages.<br>1.2 Features of JavaScript.<br>1.3 Values and Variables.<br>1.4 Operators: Assignment, Comparison, Bit wise, logical, string, special<br>1.5 Conditional statement: If statement, if..else, if..elseif, nested if, Switch..case statement.<br>1.6 Loop Statement: for loop, |

| Unit                                                   | Major Learning Outcomes<br>(in cognitive domain)                                                                                                                                | Topics and Sub-topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                        |                                                                                                                                                                                 | for...in loop, while loop,<br>do...while loop.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Unit-II<br/>Array,<br/>Function and<br/>String</b>  | 2a. Create an array in JavaScript.<br>2b. Implement a function in JavaScript.<br>2c. Perform String manipulations.                                                              | 2.1 Array: Declaring an Array, initializing Array, defining Array elements, looping in Array, Adding element in Array, sorting Array elements, combining Array elements into string, changing elements of array<br>2.2 Function: Defining a Function, writing a Function, adding arguments, scope of variables and arguments, Calling a Function, Function calling another Function, returning a value from Function.<br>2.3 String: Manipulate a String, Joining Strings, Dividing Text, copying a Substring, Converting Numbers and Strings, Numbers to Strings, Changing the Case of the String |
| <b>Unit-III<br/>Forms and<br/>Event<br/>Handling</b>   | 3a. Create a form using various elements.<br>3b. Use JavaScript to implement different form events.<br>3c. Develop JavaScript to change values dynamically for a given problem. | 3.1 Building Blocks of a Form, properties and methods of form,<br>3.2 Form Events<br>3.3 Form Objects and Elements<br>3.4 Changing Attribute Values Dynamically<br>3.5 Changing an Option List Dynamically<br>3.6 Evaluating Check Box Selections<br>3.8 Using Intrinsic JavaScript Functions<br>3.10 Disabling Elements, Read-Only Elements                                                                                                                                                                                                                                                       |
| <b>Unit-IV<br/>Cookies and<br/>Browser<br/>Windows</b> | 4a. Read and write values to cookies in JavaScript.<br>4b. Perform various operations on browser window.                                                                        | 4.1 Cookies: Creating a Cookie, Reading a Cookie, Setting the Expiration Date, Deleting a Cookie<br>4.2 Browser Window: Open the Window, Giving the New Window Focus, Placing the Window into Position on the Screen, Changing the Contents of a Window, Closing the Window, Scrolling a Web Page, Opening Multiple Windows at Once.                                                                                                                                                                                                                                                               |
| <b>Unit-V<br/>Menus and<br/>Webpage<br/>protection</b> | 5a. Develop JavaScript to implement appropriate menu as per requirement.<br>5b. Give the required security to a webpage in JavaScript                                           | 5.1 Menu: Creating a Pull-Down Menu, Dynamically Changing a Menu, Validating Menu Selections, Floating Menu, Chain Select Menu, Tab Menu, Popup Menu, Highlighted Menu, Folding Tree Menu, Microsoft Outlook Bar Style Menu, Context Menu, Scrollable Menu, Side Bar Menu, Slide-In Menu                                                                                                                                                                                                                                                                                                           |

| Unit | Major Learning Outcomes<br>(in cognitive domain) | Topics and Sub-topics                                                                                                   |
|------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
|      |                                                  | 5.2 Webpage protection:<br>Hiding Code, Disabling the Right Mouse Button, Hiding JavaScript, Concealing E-mail Address. |

\* Theory part should be covered in practical sessions.

## 5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Not Applicable

## 6.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills (**Outcomes in cognitive, psychomotor and affective domain**) so that students are able to acquire the desired programme outcome/course outcome.

**Note:** Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

| Sr. No. | Unit No. | Laboratory Learning Outcome                                            | Laboratory Experiment Title                                                                | Hours |
|---------|----------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------|
| 01      | I        | 1a. Implement different types of operators                             | • Write a JavaScript to implement different types of operators.                            | 04    |
| 02      | I        | 2a. Implement various conditional statements                           | • Write a JavaScript to implement various conditional statements.                          | 04    |
| 03      | I        | 3a. Implement various looping statements                               | • Write a JavaScript to implement various looping statements                               | 04    |
| 04      | II       | 4a. Implement concept of built in functions and user defined functions | • Write a JavaScript to implement concept of built in functions and user defined functions | 04    |
| 05      | II       | 5a. Implementation of Array functionalities.                           | • Write a JavaScript to implement Array.                                                   | 04    |
| 06      | II       | 6a. Implementation of String functionalities.                          | • Write a JavaScript to implement Strings.                                                 | 04    |
| 07      | III      | 7a. Creation of webpage using form elements.                           | • Create a webpage using form elements.                                                    | 04    |
| 08      | III      | 8a. Implement different form events on a webpage.                      | • Create a webpage to implement different form events.                                     | 04    |
| 09      | IV       | 9a. Perform operation on cookies.                                      | • Develop a webpage to perform operation on cookies.                                       | 02    |
| 10      | IV       | 9b. Implement Intrinsic Java Functions.                                | • Develop a webpage using Intrinsic Java Functions.                                        | 02    |
| 11      | IV       | 10a. Understanding browser window functionalities.                     | • Develop a webpage for placing a window on the screen and working with child window.      | 02    |
| 12      | V        | 11a. Implement validation of form fields using regular expression.     | • Develop a webpage for validation of form fields using regular expression.                | 04    |
| 13      | V        | 12a. Implement Rollover effects.                                       | • Create a webpage with Rollover                                                           | 02    |

| Sr. No.      | Unit No. | Laboratory Learning Outcome                                | Laboratory Experiment Title                            | Hours     |
|--------------|----------|------------------------------------------------------------|--------------------------------------------------------|-----------|
|              |          |                                                            | effects.                                               |           |
| 14           | V        | 13a. Develop a webpage and implement menus.                | • Create a webpage to implement menus.                 | 04        |
| 15           | V        | 14a. Develop a webpage and implement status bars.          | • Create a webpage to implement Status bars.           | 04        |
| 16           | V        | 16a. Develop a webpage and implement webpage protection.   | • Develop a webpage to implement webpage protection.   | 04        |
| 17           | V        | 15a. Develop a webpage and implement slideshow and banner. | • Develop a webpage to implement slideshow and banner. | 04        |
| <b>Total</b> |          |                                                            |                                                        | <b>60</b> |

**7.0 SELF LEARNING: (Assignment / Activities for Specific Learning / Skill Development / Online Courses / Micro projects.**  
**Not Applicable**

**8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):**

1. Arrange a lecture of a person with expertise in Javascript.
2. Use of online platforms like MOOCs, Swayam, Spoken tutorial, NPTEL, Infosys springboard etc.
3. Use any open office tools for laboratory practices.

**9.0 ASSESSMENT METHODOLOGY**

**Formative assessment (Assessment for Learning)**

Lab performance, Assignment.

**10.0 LEARNING RESOURCES:**

**A) Books:**

| Sr.No. | Author                                       | Title of Book | Publication |
|--------|----------------------------------------------|---------------|-------------|
| 1      | JavaScript Demystified                       | Keogh, Jim    | McGraw Hill |
| 2      | Beginning JavaScript                         | Wilton, Paul  | Wiley India |
| 3      | JavaScript, A Beginner's Guide Third Edition | John Pollock  | McGraw Hill |

**B) Software/Learning Websites:**

1. <https://www.w3schools.com>
2. <https://github.com>
3. <https://www.javatpoint.com>
4. <https://www.nptelvideos.com>
5. <https://www.tutorialspoint.com>

**C) Major Equipment/ Instrument with Broad Specifications:**

| Sr.No. | Equipment        | Specification                                                                                                                                                                                                               |
|--------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Desktop Computer | PC Specifications to be followed:<br>Processor: i3 or i5 (or Higher)<br>RAM: 4 GB (or Higher)<br>HDD: 1 TB SATA<br>Monitor: TFT LCD<br>OS: Genuine Windows 8 or 10 Professional or Home Premium or Windows 8 or 10 Ultimate |

|   |               |                                                |
|---|---------------|------------------------------------------------|
|   |               | Internet Connection                            |
| 2 | LCD Projector | Display Type: LCD<br>Light Output: 3200 Lumens |
| 3 | Software      | Office Suite                                   |

#### 11.0 MAPPING MATRIX OF PO's, CO's and PSO's:

| Course Outcomes | Programme Outcomes (PO's) |   |   |   |   |   |   | Programme Specific Outcomes (PSO's) |   |   |
|-----------------|---------------------------|---|---|---|---|---|---|-------------------------------------|---|---|
|                 | 1                         | 2 | 3 | 4 | 5 | 6 | 7 | 1                                   | 2 | 3 |
| CO1             | M                         | H |   |   |   |   | M | L                                   |   |   |
| CO2             |                           | H | H |   |   | L |   | L                                   | M |   |
| CO3             |                           | M |   | L | L |   |   |                                     | L | H |
| CO4             |                           | H |   |   | L | M | M | L                                   |   | M |
| CO5             | L                         | H | M |   |   |   |   |                                     |   |   |

**H: High Relationship, M: Moderate Relationship, L: Low Relationship.**

#### 12.0 SUGGESTED QUESTION PAPER PROFILE:

**Not Applicable**